Optimization and Numerical Methods

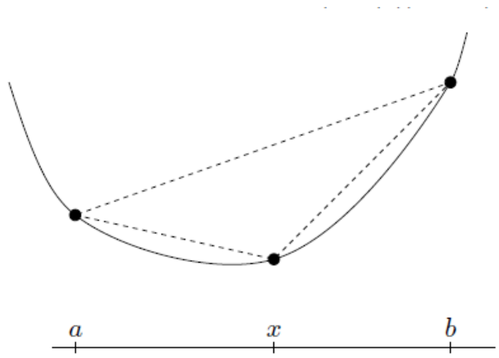
Weekly assignment answers

1 Solution set of Assignment:2 Topics: Convex function

1. Suppose $g : \mathbb{R} \to \mathbb{R}$ is convex, and c and d belongs to the domain of g with $c < d$

   a. Show that
   $$g(x) \leq \frac{d-x}{d-c} g(c) + \frac{x-c}{d-c} g(d)$$
   for all $x \in [c,d]$.
   Solution. This is Jensen's inequality with $\lambda = (d-x)/(d-c)$.

   b. Show that
   $$\frac{g(x)-g(c)}{x-c} \leq \frac{g(d)-g(c)}{d-c} \leq \frac{g(d)-g(x)}{d-x}$$
   for all $x \in (c,d)$. Draw the sketch to illustrate this inequality: determine the slopes of the lines joining the points (for example between $(x,g(x))$, $(c,g(c))$, etc) for each inequalities and compare them with each other.
   Solution. We obtain the first inequality by subtracting $g(c)$ from both sides of the inequality in (a). The second inequality follows from subtracting $g(d)$. Geometrically, the inequalities mean that the slope of the line segment between $(c,g(c))$ and $(d,g(d))$ is larger than the slope of the segment between $(c,g(c))$ and $(x,g(x))$, and smaller than the slope of the segment between $(x,g(x))$ and $(d,g(d))$.



2. For each of the following functions, determine whether it is convex or concave or neither of them.

   a. $f(x) = e^x - 1$ on **R**.
   Solution. Since the second derivative of the function (e^x) is positive for all $x \in R$. Therefore, by using the second derivative test, we can conclude that it is a convex function.

   b. $f(x_1,x_2) = 1/(x_1x_2)$ on $\mathbb{R}^2_{++}$
   Solution. The given function is a function of two variables. We use the Hessian criterion to check convexity. A function is convex if its Hessian matrix is positive semi-definite over the given domain.
   First step: compute the First-Order Partial Derivatives.
   Second step: compute the Second-Order Partial Derivatives.
   Third step: Hessian of f will be
   $$\nabla^2 f(x) = \frac{1}{x_1x_2} \begin{bmatrix} 2/x_1^2 & 1/(x_1x_2) \\ 1/(x_1x_2) & 2/x_2^2 \end{bmatrix}$$
   Fourth step: check the positive semi-definiteness of $\nabla^2 f(x)$. As all the principal minors of the matrix are positive, we conclude that the matrix is positive semi-definite. Therefore, f is convex.

3. Log-convexity and log-concavity:

   a. Show that Logistic function $f(x) = e^x/(1+e^x)$ is log-concave with domain of $f = \mathbf{R}$.

   Solution. To show that the logistic function $f(x) = e^x/(1+e^x)$ is log-concave R, we need to check whether its logarithm is concave. This means whether

$$g(x) = \log\left(\frac{e^x}{(1+e^x)}\right)$$

   is concave in R. This means we need to check whether its second derivative is non-positive. The second derivative is turns out to be

$$g''(x) = \frac{-e^x}{(1+e^x)^2}.$$

   Since e^x is positive for all $x \in R$, it implies that $g''(x) < 0$ for all $x \in \mathbb{R}$.

   The function $g(x) = \log f(x)$ is concave. Thus, the original function $f(x) = e^x/(1+e^x)$ log-concave.

   b. Show that the cumulative distribution function of the standard normal distribution is log-concave. Use the condition that a function f is log-concave if and only if $f''(x)f(x) \le (f'(x))^2$. Additional reference: section 3.5 of Boyd's book.

   Solution. (a) First we verify that $f''(x)f(x) \le (f'(x))^2$ for $x \ge 0$.

   The derivatives of f are

$$f'(x) = e^{-x^2/2}/\sqrt{2\pi}; \quad f''(x) = -xe^{-x^2/2}/\sqrt{2\pi}.$$

   So, $f''(x) \le 0$ for $x \ge 0$ and thus $f''(x)f(x) \le (f'(x))^2$ for $x \ge 0$.